

EXPERIMENT 3**TITLE: DESCRIPTIVE STATISTICS- MEASURES OF CENTRAL TENDENCY AND VARIABILITY****PROBLEM STATEMENT: -**

Perform the following operations on any open source dataset (e.g., data.csv)

1. Provide summary statistics (mean, median, minimum, maximum, standard deviation) for a dataset (age, income etc.) with numeric variables grouped by one of the qualitative (categorical) variable. For example, if your categorical variable is age groups and quantitative variable is income, then provide summary statistics of income grouped by the age groups. Create a list that contains a numeric value for each response to the categorical variable.
2. Write a Python program to display some basic statistical details like percentile, mean, standard deviation etc. of the species of 'Iris-setosa', 'Iris-versicolor' and 'Iris- versicolor' of iris.csv dataset.

Provide the codes with outputs and explain everything that you do in this step.

OBJECTIVE: To analyze and demonstrate knowledge of statistical data analysis techniques for decision-making

PREREQUISITE:-

- 1 Basic of Python Programming
- 2 Concept of statistics mean, median, minimum, maximum, standard deviation

THEORY:

The data are summarized in some, but not all ways. We chose descriptives that are either most often reported, or most often covered in introductory courses. These are as follows:

- **Central Tendency**
 - Mean
 - Median
 - Mode
- **Dispersion**
 - Standard deviation (Std. deviation)
 - Minimum
 - Maximum

- **Central Tendency**

The Mean, Median and Mode are the three measures of central tendency. Mean is the arithmetic average of a data set. This is found by adding the numbers in a data set and dividing by the number of observations in the data set. The median is the middle number in a data set when the numbers are listed in either ascending or descending order. The mode is the value that occurs the most often in a data set and the range is the difference between the highest and lowest values in a data set.

The Mean

$$\bar{x} = \frac{\sum x}{N}$$

Here,

$\sum$ represents the summation

X represents observations

N represents the number of observations .

In the case where the data is presented in a tabular form, the following formula is used to compute the mean

$$\text{Mean} = \sum f x / \sum f$$

Where $\sum f = N$

The Median

If the total number of observations (n) is an odd number, then the formula is given below:

$$\text{Median} = \left(\frac{n+1}{2} \right)^{th} \text{ observation}$$

If the total number of the observations (n) is an even number, then the formula is given below:

$$\text{Median} = \frac{\left(\frac{n}{2} \right)^{th} \text{ observation} + \left(\frac{n}{2} + 1 \right)^{th} \text{ observation}}{2}$$

Consider the case where the data is continuous and presented in the form of a frequency distribution, the median formula is as follows.

Find the median class, the total count of observations $\sum f$.

The median class consists of the class in which (n / 2) is present.

$$\text{Median} = l + \left[\frac{\frac{n}{2} - c}{f} \right] \times h$$

Here

l = lesser limit belonging to the median class

c = cumulative frequency value of the class before the median class

f = frequency possessed by the median class

h = size of the class

The Mode

The mode is the most frequently occurring observation or value.

Consider the case where the data is continuous and the value of mode can be computed using the following steps.

- Determine the modal class that is the class possessing the maximum frequency.
- Calculate the mode using the below formula

$$\text{Mode} = l + \left[\frac{f_m - f_1}{2f_m - f_1 - f_2} \right] \times h$$

l = lesser limit of modal class

f_m = frequency possessed by the modal class

f_1 = frequency possessed by the class before the modal class

f_2 = frequency possessed by the class after the modal class

h = width of the class

Standard deviation

$$s = \sqrt{\frac{\sum (X - \bar{x})^2}{n - 1}}$$

- s = sample standard deviation
- Σ = sum of...
- X = each value
- $\bar{x}$ = sample mean
- n = number of values in the sample

Steps for calculating the standard deviation

The standard deviation is usually calculated automatically by whichever software you use for your statistical analysis. But you can also calculate it by hand to better understand how the formula works.

There are six main steps for finding the standard deviation by hand. We'll use a small data set of 6 scores to walk through the steps.

Data set					
46	69	32	60	52	41

1. Step 1: Find the mean

To find the mean, add up all the scores, then divide them by the number of scores.

Mean ($\bar{x}$)
$\bar{x} = (46 + 69 + 32 + 60 + 52 + 41) \div 6 = 50$

2. Step 2: Find each score's deviation from the mean

Subtract the mean from each score to get the deviations from the mean.

Since $\bar{x} = 50$, here we take away 50 from each score.

Score	Deviation from the mean
46	$46 - 50 = -4$
69	$69 - 50 = 19$
32	$32 - 50 = -18$
60	$60 - 50 = 10$

Score	Deviation from the mean
52	$52 - 50 = 2$
41	$41 - 50 = -9$

3. Step 3: Square each deviation from the mean

Multiply each deviation from the mean by itself. This will result in positive numbers.

Squared deviations from the mean
$(-4)^2 = 4 \times 4 = 16$
$19^2 = 19 \times 19 = 361$
$(-18)^2 = -18 \times -18 = 324$
$10^2 = 10 \times 10 = 100$
$2^2 = 2 \times 2 = 4$
$(-9)^2 = -9 \times -9 = 81$

4. Step 4: Find the sum of squares

Add up all of the squared deviations. This is called the sum of squares.

Sum of squares
$16 + 361 + 324 + 100 + 4 + 81 = 886$

5. Step 5: Find the variance

Divide the sum of the squares by $n - 1$ (for a sample) or N (for a population) – this is the variance.

Since we're working with a sample size of 6, we will use $n - 1$, where $n = 6$.

Variance
$886 \div (6 - 1) = 886 \div 5 = 177.2$

6. Step 6: Find the square root of the variance

To find the standard deviation, we take the square root of the variance.

Standard deviation

$$\sqrt{177.2} = 13.31$$

From learning that $SD = 13.31$, we can say that each score deviates from the mean by 13.31 points on average.

Consider HR dataset it contains fields like Age, MonthlyIncome, Attrition, BusinessTravel and so on so following are steps to find statistics (mean, median, minimum, maximum, standard deviation)

#Mean of monthly income and age

```
print("The mean of monthly income is :", df.loc[:, "MonthlyIncome"].mean())
print("The mean of age is :", df.loc[:, "Age"].mean())
```

#Mode of monthly income and age

```
print("The median of monthly income is :", df.loc[:, "MonthlyIncome"].median())
print("The median of age is :", df.loc[:, "Age"].median())
```

#Median of monthly income and age

```
print("The mode of monthly income is :", df.loc[:, "MonthlyIncome"].mode())
print("The mode of age is :", df.loc[:, "Age"].mode())
```

#Standard deviation of monthly income and age

```
print("The standard deviation of monthly income is :", df.loc[:, "MonthlyIncome"].std())
print("The standard deviation of age is :", df.loc[:, "Age"].std())
```

#Storing age and monthly income in array and then finding maximum and minimum values

```
array1 = np.array(df["MonthlyIncome"])
array2 = np.array(df["Age"])
print("Income", array1)
print("Age array", array2)
```

```

print("Maximum income among the employees is :",max(array1))
print("Minimum income among the employees is :",min(array1))
print("Maximum age among the employees is :",max(array2))
print("Minimum age among the employees is :",min(array2))

# Replacing the categorical values by numeric values
df.head()
df["BusinessTravel"].replace({"Travel_Rarely":1, "Travel_Frequently":0}, inplace=True)
df["Attrition"].replace({"Yes":1, "No":0}, inplace=True)
df.head()

```

##Use of describe()

To display basic statistical details like percentile, mean, standard deviation etc. for Iris-Virginica use describe

Iris-setosa				
	Sepal_Length	Sepal_Width	Petal_Length	Petal_Width
count	50.00000	50.000000	50.000000	50.000000
mean	5.00600	3.418000	1.464000	0.24400
std	0.35249	0.381024	0.173511	0.10721
min	4.30000	2.300000	1.000000	0.10000
25%	4.80000	3.125000	1.400000	0.20000
50%	5.00000	3.400000	1.500000	0.20000
75%	5.20000	3.675000	1.575000	0.30000
max	5.80000	4.400000	1.900000	0.60000
Iris-versicolor				
	Sepal_Length	Sepal_Width	Petal_Length	Petal_Width
count	50.000000	50.000000	50.000000	50.000000
mean	5.936000	2.770000	4.260000	1.326000
std	0.516171	0.313798	0.469911	0.197753
min	4.900000	2.000000	3.000000	1.000000
25%	5.600000	2.525000	4.000000	1.200000
50%	5.900000	2.800000	4.350000	1.300000
75%	6.300000	3.000000	4.600000	1.500000
max	7.000000	3.400000	5.100000	1.800000
Iris-virginica				
	Sepal_Length	Sepal_Width	Petal_Length	Petal_Width
count	50.00000	50.000000	50.000000	50.000000
mean	6.58800	2.974000	5.552000	2.02600
std	0.63588	0.322497	0.551895	0.27465
min	4.90000	2.200000	4.500000	1.40000
25%	6.22500	2.800000	5.100000	1.80000
50%	6.50000	3.000000	5.550000	2.00000
75%	6.90000	3.175000	5.875000	2.30000
max	7.90000	3.800000	6.900000	2.50000

Conclusion: In this way we have explored the functions of the python library for to calculate Data statistics (mean, median, minimum, maximum, standard deviation).

Assignment Question:

1. **How to find Mean Mode and Median.**
2. **How to find standard deviation.**
3. **Write a Python program to display some basic statistical details like percentile, mean, standard deviation etc.**
4. **What is use of Describe() Command?**